Gravity

Building Models to Describe our World

Kepler, Newton, Gauss and Einstein

Outline

Keplerian and Newtonian Gravity

Building Models to Describe our World

Kepler, Newton, Gauss and Einstein

Kepler's Laws

1. The path of a planet around the Sun is described by an ellipse with the Sun at one of its foci.
2. All planets move in such a way that the area swept by a line connecting the planet and the Sun in a given period of time is constant.
3. The ratio between the orbital periods, T, of two planets squared is equal to the ratio of the semi-major axes, s, of their orbits cubed:

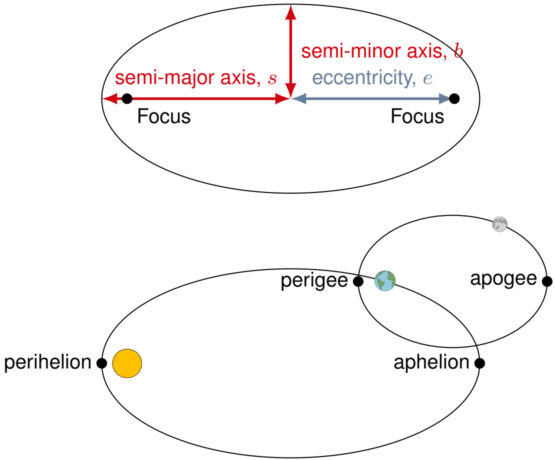
$$\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{s_1}{s_2}\right)^3$$

Kepler's First Law

- Orbits of planets are ellipses (but some bodies that are not in bound orbits will have parabolic or hyperbolic orbits).
- Ellipses are characterized by the location of two foci and one dimension (e.g. semi-major/minor axis, eccentricity).

$$\frac{x^2}{s^2} + \frac{y^2}{b^2} = 1$$

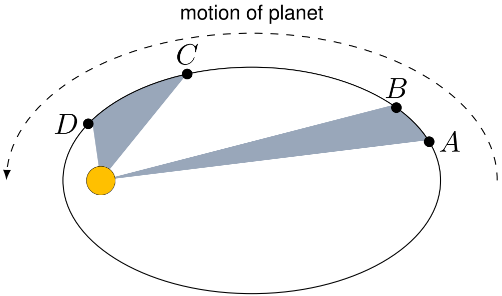
$$e = \sqrt{s^2 - b^2} \quad (s \geq b)$$



Elliptical orbits and their parameters.

Kepler's Second Law

- "Equal areas swept out in equal times":
  - → Planets move faster near the perihelion.
  - → Bodies in a circular orbit move at constant speed.



Planets sweep out equal areas in equal amounts of time.

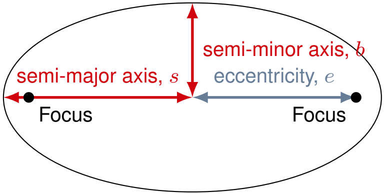
Kepler's Third Law

- The ratio between the orbital periods, T, of two planets squared is equal to the ratio of the semi-major axes, s, of their orbits cubed:

$$\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{s_1}{s_2}\right)^3$$

- It's a mouthful... But it just means that:

$$\frac{T^2}{s^3} = \text{constant}$$



Kepler's Third Law relates to the semi-major axis.

That constant depends on the mass of the body that the planets orbit (e.g. the Sun). Thus, by measuring T and s, you can determine the mass of the Sun!

Newton's Universal Theory of Gravity (magnitude)

- Two bodies with mass will exert an **attractive** force on each other that is proportional to the **product of their masses** and inversely proportional to the **distance between them squared**:

$$F \propto \frac{M_1 M_2}{r^2}$$

- The constant of proportionality is called the gravitational constant, G:

$$F = G\frac{M_1 M_2}{r^2}$$

The Cavendish experiment

- Want to measure G.
- The forces involved are minuscule (less than 1×10^{-6} Newtons).
- Clever use of a “torsion pendulum”: two masses suspended by a thin fibre.
- Bring two big masses close to the suspended masses and measure by how much they twist.
- Very sensitive experiment...

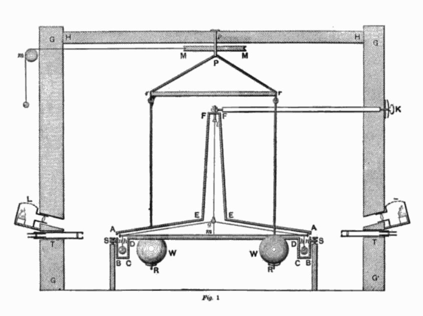


Figure from Cavendish’s paper showing the torsion pendulum and building it was housed in. The large masses could be rotated from the outside through a system of pulleys. Philosophical Transactions of the Royal Society of London, (part II) ***88*** *p.469-526 (21 June 1798).*

Newton's Universal Theory of Gravity (magnitude and direction)

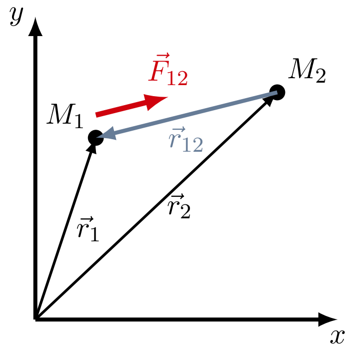
The force, $\vec{F}_{12}$, **on** mass M_1 **from** M_2 can be written as:

$$\vec{F}_{12} = -G\frac{M_1 M_2}{r_{12}^2}\hat{r}_{12} = -G\frac{M_1 M_2}{r_{12}^{3}}\vec{r}_{12}$$

where:

$$\vec{r}_{12} = \vec{r}_1 - \vec{r}_2 \;\; \text{and} \;\; \hat{r}_{12} = \frac{\vec{r}_{12}}{r_{12}}$$

In general, we use the **minus sign to indicate that the force is attractive** (we could've used a plus sign and the vector $\vec{r}_2 - \vec{r}_1$, but we don't!).

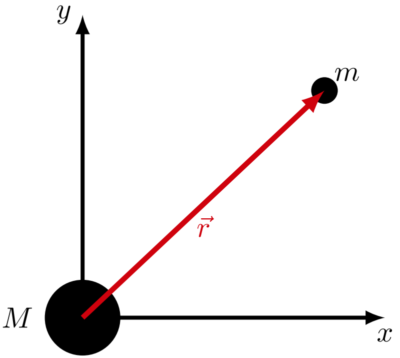


Vectors in Newton's Universal Theory of Gravity. The force on M_1 is in the opposite direction of $\vec{r}_{12}$.

When one mass is much larger

- If M is much bigger than m, M can be considered as effectively stationary (e.g. the Sun when considering the motion of planets, the Earth when considering the motion of satellites).
- Makes sense to make M the origin of our coordinate system, so the force on m (at position, $\vec{r}$ is):

$$\vec{F} = -G\frac{Mm}{r^2}\hat{r} = -G\frac{Mm}{r^3}\vec{r}$$

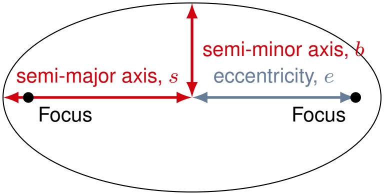


Newton's Universal Theory of Gravity with one mass at the origin.

The constant from Kepler's Third Law

- The constant is given by:

$$\frac{T^2}{s^3} = \text{constant}$$



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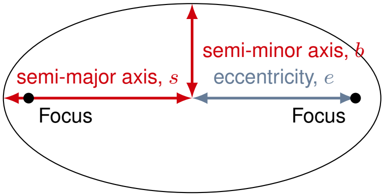
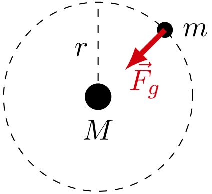


- Can evaluate this using a circular orbit:

$$G\frac{Mm}{r^2} = m\frac{v^2}{r} \quad \rightarrow \quad v = \sqrt{\frac{GM}{r}}$$

$$v = \frac{2\pi r}{T} \quad \rightarrow \quad T = \frac{2\pi r}{v} = \frac{2\pi r}{\sqrt{\frac{GM}{r}}}$$



$$\sum F = F_g = m\frac{v^2}{r}$$

For a circular orbit, the semi-major axis is just the radius. The force of gravity is responsible for the radial acceleration.

The constant from Kepler's Third Law

- The constant is given by:

$$\frac{T^2}{s^3} = \text{constant}$$

- Can evaluate this using a circular orbit:

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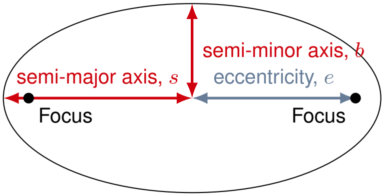
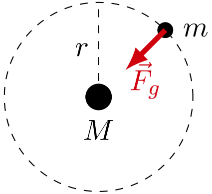
$$v = \frac{2\pi r}{T} \quad \rightarrow \quad T = \frac{2\pi r}{v} = \frac{2\pi r}{\sqrt{\frac{GM}{r}}}$$

- The constant is then given by:

$$\frac{T^2}{r^3} = \frac{4\pi^2 r^2}{r^3 \frac{GM}{r}} = \frac{4\pi^2}{GM}$$





$$\sum F = F_g = m\frac{v^2}{r}$$

For a circular orbit, the semi-major axis is just the radius. The force of gravity is responsible for the radial acceleration.

The Gravitational Field and Gauss’ Law

Building Models to Describe our World

Kepler, Newton, Gauss and Einstein

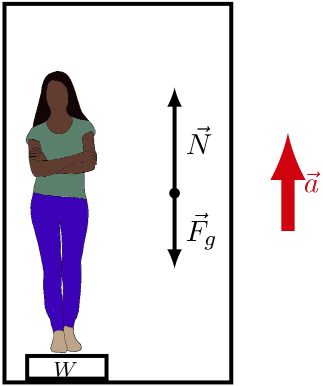
Apparent weight

- Can be defined as **the net force ($\sum \vec{F}$) on you minus the force of gravity**.
- If you stand on a scale, the only 2 forces exerted on you are gravity and the normal force:
  - → The normal force is your apparent weight:

$$\sum \vec{F} = \vec{F}_g + \vec{N}$$

$$\vec{W} = \sum \vec{F} - \vec{F}_g = \vec{N}$$

  - → The scale measures the magnitude of the normal force.
- You feel “weightless” when your apparent weight is zero, **not** when you have no weight.



A person on a scale in an elevator accelerating upwards. The scale reads the value of the apparent weight, which is equal to the normal force.

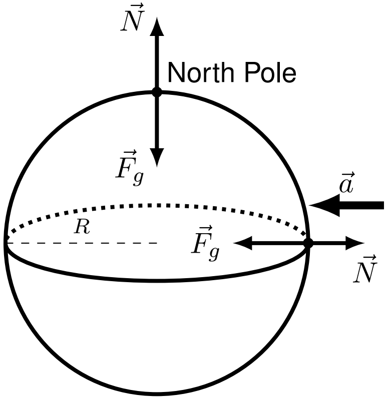
Apparent weight on Earth

- At the North Pole, you are not accelerating, so the normal force has the same magnitude as your weight.
- At the equator, you are accelerating towards the centre of the Earth (uniform circular motion) – the normal force is thus smaller,
- The effect is small...

$$\vec{W} = \sum \vec{F} - \vec{F}_g = \vec{N}$$

$$\sum F_r = F_g - N = m\frac{v^2}{R}$$

$$W = N = mg = mg - m\frac{v^2}{R} = m\left(g - \frac{v^2}{R}\right)$$



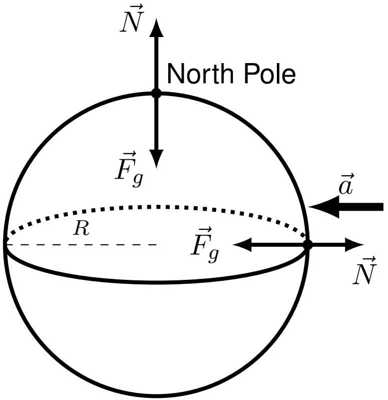
Apparent weight on a rotating Earth. The normal force is smaller at the equator since some of the weight contributes to the centripetal acceleration.

Apparent weight on Earth

- Your apparent weight depends on how fast you rotate about the centre of the Earth:

$$W = m\left(g - \frac{v^2}{R}\right)$$

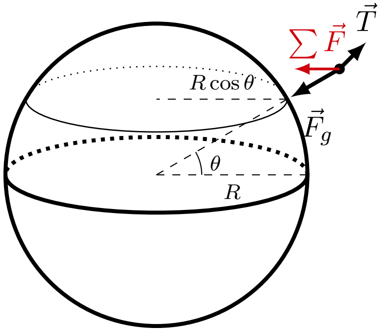
- If you go just the right speed, your apparent weight will be zero:
- This is what being "weightless" means.
- If your acceleration is g, then you feel weightless. If the only force exerted on you is that from gravity, you feel weightless.



Apparent weight on a rotating Earth. The normal force is smaller at the equator since some of the weight contributes to the centripetal acceleration.

A plumb line

- We use plumb lines to determine the vertical direction.
- The mass on the plumb line accelerates with the Earth, so the net force on it is not zero (and the tension is thus not opposite of gravity). The line does not point to the centre of the Earth (except at Pole/Equator).
- If you cut the line, it's a little complicated, since the mass does not have an initial speed of zero (as it rotates around the Earth) http://articles.adsabs.harvard.edu/pdf/1897PA.....5..186W
- The effect is very small for the Earth.



Except at the poles and equator, a plumb line does not point towards the centre of the Earth!

Gravitational field

- Near the centre of the Earth, we use the force of gravity as:

$$\vec{F}_g = m\vec{g}$$

- We call, $\vec{g}$, the **gravitational field** of the Earth (rather than the acceleration due to gravity).
- This has to be the same as the force from Newton's Theory of Gravity:

$$F_g = G\frac{Mm}{R^2}$$

- Thus, in terms of the mass (M) and radius (R) of the Earth, the gravitational field at the surface of the Earth is:

$$g = \frac{F_g}{m} = G\frac{M}{R^2}$$

Gravitational field, $\vec{g}$

- The force on m from M:

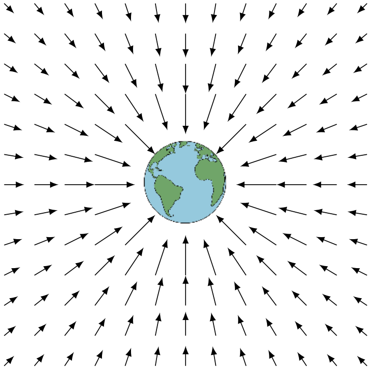
$$\vec{F}(\vec{r}) = -G\frac{Mm}{r^2}\hat{r}$$

- The force from M **per unit mass** is called the gravitational field of M:

$$\vec{g}(\vec{r}) = \frac{\vec{F}(\vec{r})}{m} = -G\frac{M}{r^2}\hat{r}$$

- The gravitational field is associated with M, and makes it easy to calculate the force from M on any mass, m':

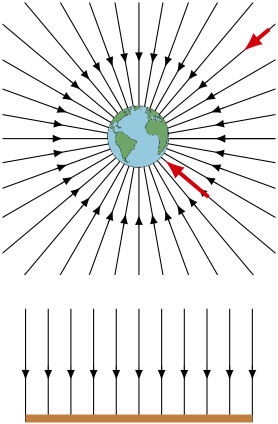
$$\vec{F}'(\vec{r}) = m'\vec{g}(\vec{r})$$



Gravitational field vectors around the Earth.

Visualizing the gravitational field: field lines

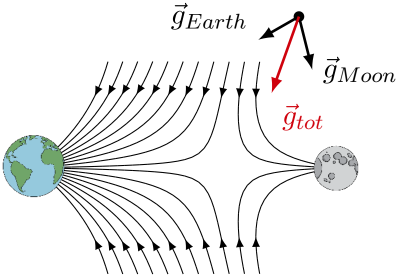
- If you have a mass M in space, one can think of the gravitational field as existing everywhere in space around M.
- We can visualize this with **field lines**.
- The gravitational field vector is in the **direction tangent to the field line** and has a **magnitude proportional to the density of field lines**.
- Near the surface of the Earth, these are parallel lines downwards.



Gravitational field lines around the Earth. Close to the surface, they are effectively parallel, so the field is essentially constant.

The Superposition Principle

- If there are two bodies, say M_1 and M_2, we can calculate the field from each body separately.
- The total gravitational field at some point in space is then the sum of the individual gravitational fields.
- If you calculate the net force on a satellite from both the Earth and Moon, you would add the force vectors. Since the g-field is just the force per unit mass, you can add those together as well.
- The field from the Moon does not depend on the field from the Earth (superposition).

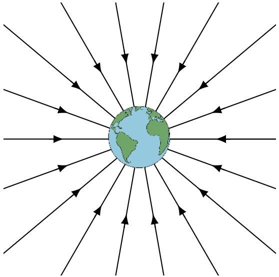


Gravitational field lines for an Earth-Moon system.

Gauss' Law

- Gauss' Law allows us to calculate the magnitude of the gravitational field in situations with high symmetry.
- **Newton's Universal Law of Gravity is only valid for point masses** (most masses, like the Earth, are not a point mass) – How would you calculate the gravitational field/force at a location half-way inside the Earth?
- It is a consequence of Gauss' Law that we can model the Earth as a point mass, with all the mass at the centre of the Earth, if we want to calculate the force of gravity/gravitational field **outside** of the Earth.

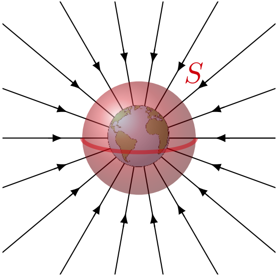
Gauss' Law - The basic idea



Gravitational field lines around the Earth.

- If you **enclose** a mass with a surface, S, then the number of field lines that cross that surface depends on the mass that you've enclosed.

Gauss' Law - The basic idea



Gravitational field lines around the Earth. Enclosing the Earth with a gaussian spherical surface, S.

- If you **enclose** a mass with a surface, S, then the number of field lines that cross that surface depends on the mass that you've enclosed.
- Makes sense, if you have a heavier mass, you have a denser field lines, so more of them will enter S.

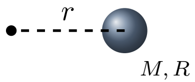
Gauss' Law

$$\oint \vec{g}(\vec{r}) \cdot d\vec{A} = 4\pi G M^{enc}$$

- Two sides to the equation:
  1. Left side: **flux of the field through the surface.** It is a measure of the number of field lines through the surface.
  2. Right-side: amount of **mass enclosed by the surface** that *we chose*.
- The flux is a "surface integral" over a closed surface. It is the sum of the dot products of the field, $\vec{g}(\vec{r})$, with a vector, $d\vec{A}$, that is always normal to the surface.
- To use Gauss' Law:
  1. Choose a closed surface and figure out how much mass is inside.
  2. Figure out how much flux is coming out the surface.
  3. Set them equal and determine, say, the gravitational field.
- Gauss' Law is most useful in situations with a high degree of symmetry (e.g. spherical symmetry).

Field outside of a uniform sphere of radius, R, and mass, M

Want the field, $\vec{g}(\vec{r})$, at some distance r from the centre:

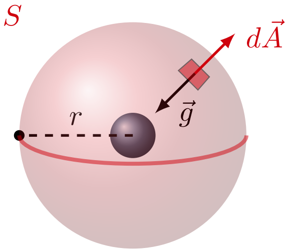


Gauss' Law to find field outside of a spherical mass of radius, R, and mass, M.

Field outside of a uniform sphere of radius, R, and mass, M

Want the field, $\vec{g}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius r.



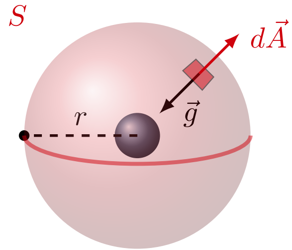
Gauss' Law to find field outside of a spherical mass of radius, R, and mass, M.

Field outside of a uniform sphere of radius, R, and mass, M

Want the field, $\vec{g}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius r.
2. Evaluate the flux integral (use symmetry arguments):

$$\oint \vec{g}(\vec{r}) \cdot d\vec{A} = -\oint g(r)dA = -g(r)\oint dA$$



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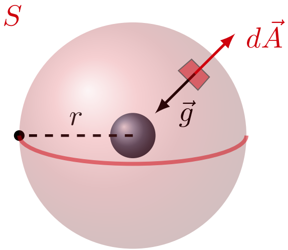
2. Evaluate the flux integral (use symmetry arguments):

$$\oint \vec{g}(\vec{r}) \cdot d\vec{A} = -\oint g(r) dA = -g(r) \oint dA$$

$\rightarrow$ Adding together all of the dA:

$$\oint dA = 4\pi r^2$$

$$\therefore \oint \vec{g}(\vec{r}) \cdot d\vec{A} = -4\pi r^2 g(r)$$



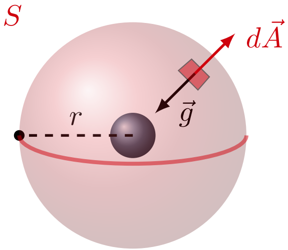
Gauss' Law to find field outside of a spherical mass of radius, R, and mass, M.

Field outside of a uniform sphere of radius, R, and mass, M

Want the field, $\vec{g}(\vec{r})$, at some distance r from the centre:

1. Choose a spherical surface (S) of radius r.
2. Evaluate the flux integral: $-4\pi r^2 g(r)$
3. Determine the mass enclosed. In this case, all of M is enclosed:

$$M^{enc} = M$$



Gauss' Law to find field outside of a spherical mass of radius, R, and mass, M.

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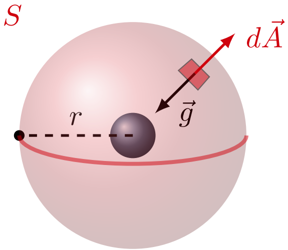
1. Choose a spherical surface (S) of radius r.
2. Evaluate the flux integral: $-4\pi r^2 g(r)$
3. Determine the mass enclosed: $M^{enc} = M$
4. Put it all together:

$$\oint \vec{g}(\vec{r}) \cdot d\vec{A} = 4\pi G M^{enc}$$

$$-4\pi r^2 g(r) = 4\pi G M$$

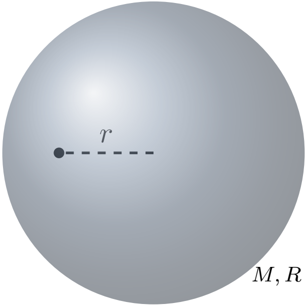
$$\therefore g(r) = -G\frac{M}{r^2}$$



Gauss' Law to find field outside of a spherical mass of radius, R, and mass, M.

Field *inside* of a uniform sphere of radius, R, and mass, M

Want the field, $\vec{g}(\vec{r})$, at some distance $r < R$ from the centre:



Gauss' Law to find field inside of a spherical mass of radius, R, and mass, M.

Field *inside* of a uniform sphere of radius, R, and mass, M

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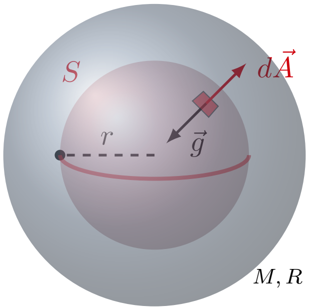


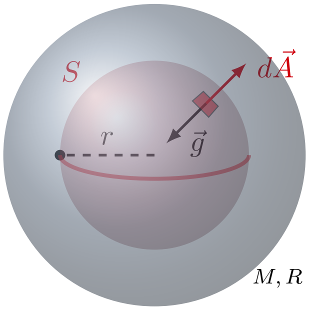
Gauss' Law to find field inside of a spherical mass of radius, R, and mass, M.

Field *inside* of a uniform sphere of radius, R, and mass, M

Want the field, $\vec{g}(\vec{r})$, at some distance $r < R$ from the centre:

1. Choose a spherical surface (S) of radius r.
2. Evaluate the flux integral (same as before):

$$\oint \vec{g}(\vec{r}) \cdot d\vec{A} = -\oint g(r) dA = -g(r) \oint dA$$
$$= -4\pi r^2 g(r)$$



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Field *inside* of a uniform sphere of radius, R, and mass, M

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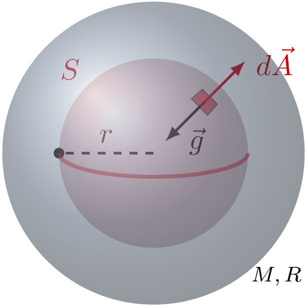
1. Choose a spherical surface (S) of radius r.
2. Evaluate the flux integral: $-4\pi r^2 g(r)$
3. Determine the mass enclosed. Use the density, $\rho = \frac{M}{V}$:

$$\rho = \frac{M}{\frac{4}{3}\pi R^3}$$

$$M^{enc} = \rho\frac{4}{3}\pi r^3 = \frac{M}{\frac{4}{3}\pi R^3}\frac{4}{3}\pi r^3$$

$$= M\frac{r^3}{R^3}$$



Gauss' Law to find field inside of a spherical mass of radius, R, and mass, M.

Field *inside* of a uniform sphere of radius, R, and mass, M

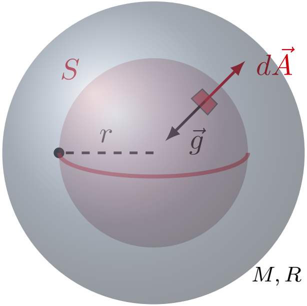
Want the field, $\vec{g}(\vec{r})$, at some distance $r < R$ from the centre:

1. Choose a spherical surface (S) of radius r.
2. Evaluate the flux integral: $-4\pi r^2 g(r)$
3. Determine the mass enclosed: $M^{enc} = M\frac{r^3}{R^3}$
4. Put it all together:

$$\oint \vec{g}(\vec{r}) \cdot d\vec{A} = 4\pi G M^{enc}$$

$$-4\pi r^2 g(r) = 4\pi G M \frac{r^3}{R^3}$$

$$\therefore g(r) = -G\frac{M}{R^3} r$$



Gauss' Law to find field inside of a spherical mass of radius, R, and mass, M.

Inside the sphere, the field increases linearly with radius. Outside it decreases with r^2.

Gravitational Potential and General Relativity

Building Models to Describe our World

Kepler, Newton, Gauss and Einstein

Gravitational potential energy

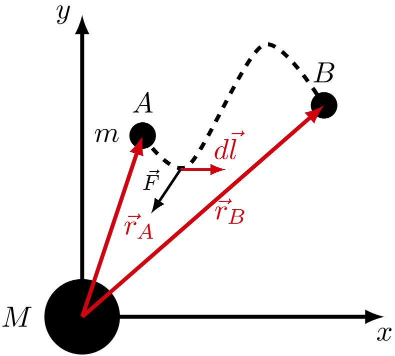
- The force exerted on mass m from mass M:

$$\vec{F} = -G\frac{Mm}{r^2}\hat{r} = -G\frac{Mm}{r^3}\vec{r}$$

- To determine the change in potential energy, we need to calculate the work done along a particular path:

$$W = \int_A^B \vec{F}\cdot d\vec{l}$$

$$\Delta U = U(\vec{r}_B) - U(\vec{r}_A)$$

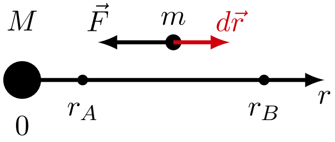


The work done by gravity on mass m in going from A to B, depends only on the end points.

Determining the gravitational potential energy function, $U(\vec{r})$

- Choose straight line as a path, to make it 1D (and our lives easier):

$$W = \int_{r_A}^{r_B} \vec{F} \cdot d\vec{r} = \int_{r_A}^{r_B} F(r) dr = \int_{r_A}^{r_B} -G\frac{Mm}{r^2} dr$$



Calculating the work done on m by gravity along a straight, radial, path, from r_A to r_B

Determining the gravitational potential energy function, $U(\vec{r})$

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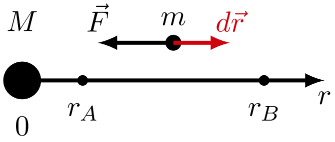
$$= \left[G\frac{Mm}{r} \right]_{r_A}^{r_B} = G\frac{Mm}{r_B} - G\frac{Mm}{r_A} = -\Delta U$$



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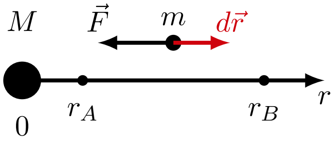


Calculating the work done on m by gravity along a straight, radial, path, from r_A to r_B

- By inspection:

$$\Delta U = G\frac{Mm}{r_A} - G\frac{Mm}{r_B} = U(r_B) - U(r_A)$$
$$\therefore U = -G\frac{Mm}{r} + C$$

Gravitational potential energy

- If m is a distance r away from M, it will have gravitational potential energy:

$$U = -G\frac{Mm}{r} + C$$

- A convenient choice of constant is $C = 0 \rightarrow$ **Assume this choice from now on**:
  $\rightarrow$ The gravitational potential energy of m becomes 0 when m is infinitely far from M ($r \rightarrow \infty$).

Mechanical energy

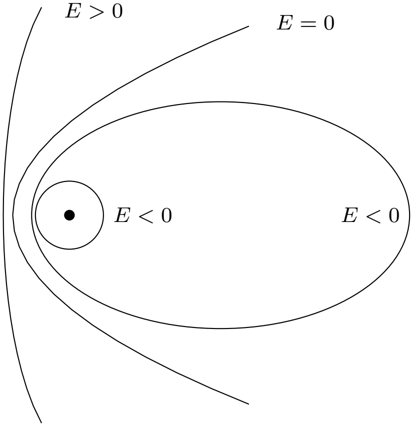
- We can define the mechanical energy of m, when it is a distance r away from M:

$$E = K + U = \frac{1}{2}mv^2 - G\frac{Mm}{r}$$

- If gravity is the only force on m, then the mechanical energy of m is conserved (constant).
- This is useful in analyzing possible orbits of m around M.
- As $r \to \infty$, the potential energy goes to zero, and all of the energy must be kinetic.

Possible orbits

1. If $E < 0$, m can never make it to infinity, as kinetic energy would be negative $\rightarrow m$ is "gravitationally bound" (as in to bind).
   - If m has the right velocity, it will be in a circular orbit.
   - Otherwise, m will be in an elliptical orbit.
2. If $E = 0$, m will just barely make it to infinity with zero kinetic energy.
   - This will result in a parabolic orbit.
   - m can "escape" the gravitational pull of M (just barely).
3. If $E > 0$, m will have a non-zero speed when it reaches infinity.
   - The orbit is a hyperbola.
   - m can escape the gravitational pull of M.



Possible orbits depending on energy.

Why GR?

- Newton's Theory does not explain the orbit of Mercury.
- Newton's Theory is **fundamentally incompatible** with Special Relativity which requires that no signal can ever travel faster than the speed of light (causality would be violated if something can travel faster than the speed of light).

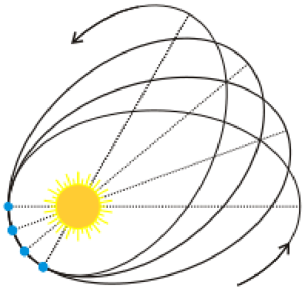


Illustration of the precession of Mercury.

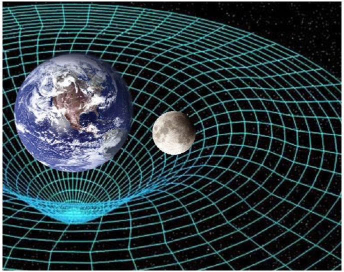
https:// en.wikipedia.org/ wiki/ Apsidal_precession

What is GR?

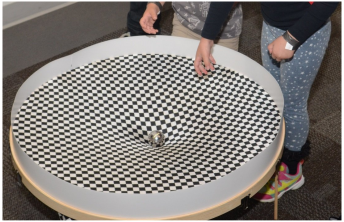
- Based on the equivalence principle:
    - There's nothing in physics that requires that the m in $F = ma$ is the same m as in $G\frac{Mm}{r^2}$.
- Really complicated math... "Differential geometry":

$$\text{(Curvature)} \quad G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad \text{(Energy)}$$

Objects are not deflected by the force of gravity. Instead, matter/energy bends space-time so that objects just follow "straight lines" in that space. Objects just move inertially (like Newton's first law) through bent space. That is the connection between inertial and gravitational mass.



Space time around Earth.

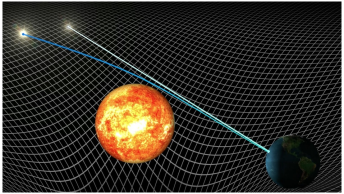


A simulation of curved space.

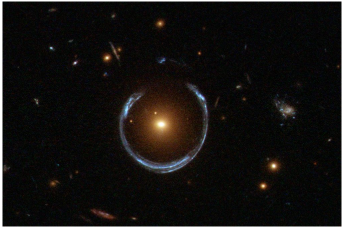
http://www.thestargarden.co.uk/General-relativity.html

Some consequences of GR

- Since light goes in a straight line, and space is curved near a massive object, light will appear to bend around massive objects.
- Time goes by slower when the gravitational field is stronger (GPS needs to take this into account as time goes by slower near the surface of the Earth than in orbit).
- The Universe itself is expanding. It's not expanding into something, space itself is expanding (the distance between stationary objects grows with time).
- Gravitational waves exist (disturbances in Space Time that propagate like ripples on a pond).



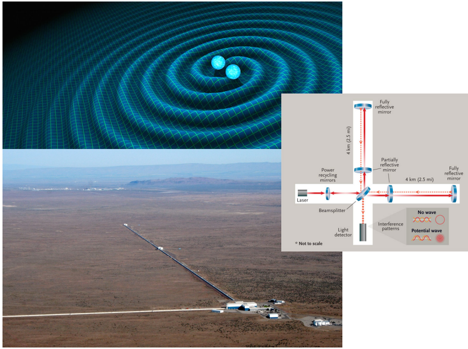
Light bending around the Sun, first successful test of GR.



Einstein rings.

Gravitational Waves

- LIGO experiment recently (2016) detected gravitational waves for the first time $\rightarrow$ 2017 Nobel Prize.
- It measures distortions in space as a gravitational wave passes by (less than one ten thousandth of the diameter of a proton on a distance of $4\,\mathrm{km}$!).
- The wave was created from an event that released a lot of gravitational potential energy (two black holes falling on each other).



Daw, E. (2016, February 11). Gravitational waves discovered: How did the experiment at LIGO actually work? The Conversation. https://theconversation.com/gravitational-waves-discovered-how-did-the-experiment-at-ligo-actually-work-54510,
Gravitational Wave Detection Heralds New Era. (n.d.). Sky & Telescope. https://skyandtelescope.org/astronomy-news/gravitational-wave-detection-heralds-new-era-of-science-0211201644

Issues with GR

- Mathematically, it is very different from the theories for the other 3 forces of nature – it's not clear how to combine it into a unified theory of physics (in particular, the "quantum" aspect).
- It has some unusual consequences:
  - Most of the energy (70%) in the Universe is a "negative energy" ("Dark energy") that is responsible for the Universe expanding $\rightarrow$ we know this from measuring the curvature of space and rate of expansion.
  - Most of the matter in the Universe is in a non-luminous form that we have never detected ("Dark Matter") $\rightarrow$ we are trying to detect it; we think it's just a different type of particle.